

the Broadband Forum. The software is publicly available with a highly permissive open source licence.”

Microsoft releases 13 billion parameter language and vision chatbot LLaVA

Microsoft has unveiled LLaVA, an advanced language and vision chatbot that boasts a whopping 13 billion parameters. This remarkable bot was developed through the collaboration of experts from Microsoft, the University of Wisconsin–Madison, and Columbia University.

To evaluate the performance of LLaVA, the researchers used GPT-4, a widely-recognised OpenAI language model. To do this, the team provided the model with synthetic conversations between humans and AI assistants. Afterwards, LLaVA’s



responses were rated on a scale of 1 to 10, where it achieved an impressive score of 8.5. When it was fine-tuned using the ScienceQA training data set, it established a new record with an accuracy of 92.53%.

According to the researchers, “This (paper) demonstrates the effectiveness of visual instruction tuning using language-only GPT-4.

We have presented an automatic pipeline to create language-image instruction-following data, based on which we train LLaVA, a multimodal model to follow human intent to complete visual tasks. It achieves [an] excellent visual chat experience when fine-tuned on multimodal chat data.”

In addition to being able to answer questions related to images, LLaVA is capable of adhering to instructions and creating pieces of text in various formats -- including poetry, code, musical compositions, emails, and letters. This revolutionary chatbot is still in its development stage, but has already showcased a range of abilities usually performed by human beings.

UAE’s Falcon 40B is the number one AI model

The Falcon 40B AI model developed by Abu Dhabi’s Technology Innovation Institute (TII) has catapulted to the number one position on Hugging Face’s Open Large Language Model (LLM)



Leaderboard. This impressive feat demonstrates the exceptional capability of the Falcon 40B, securing its place as a world leader in the domain of artificial intelligence.

Hugging Face provides the industry’s leading independent

verification for AI models. With its power and capabilities, Falcon 40B triumphed over the well-established LLaMA of Meta, Stability AI’s StableLM, and Together’s RedPajama, firmly solidifying its standing on the Leaderboard.

The evaluation system consisted of four major assessments -- the AI2 Reasoning Challenge (25-shot), HellaSwag (10-shot), MMLU (5-shot), and TruthfulQA (0-shot). The open LLM Leaderboard serves as an incredibly valuable resource for the AI community, recording, evaluating, and grading LLMs and chatbots as they are released.

Falcon 40B was trained on a remarkable one trillion tokens, reinforcing the UAE’s push to become an international powerhouse in the AI arena. TII has already initiated work on Falcon’s subsequent model, the 180B AI version.

Netreap: Cosmonic’s open source project combines Cilium and Nomad

Cosmonic, a provider of a platform-as-a-service (PaaS) for building and deploying WebAssembly (Wasm) applications, has unveiled its latest open source project: Netreap. The project amalgamates Cilium and Nomad, two popular open source projects that specialise in container networking and orchestration.

Cilium is a cloud-native application’s overlay for offering policy, security, and observability, whereas Nomad is an orchestration platform used for managing and deploying containerised applications. When both projects are joined together, Netreap becomes a robust platform for building and launching secure, robust, and reliable Wasm applications.

According to Cosmonic’s head of infrastructure, Dan Norris, “The best CNI (container networking interface) is Cilium.” However, Cilium is better adapted to Kubernetes and it was challenging to integrate Cilium with Nomad. The Netreap custom operator was designed by the Cosmonic team to deal with this specific issue.

Netreap is equipped with a Consul key to store the full JSON format of Cilium policies in the cluster, and it syncs it with the local Cilium agent. With Netreap, it’s possible to apply security policies to Wasm applications without having to depend on Kubernetes.

